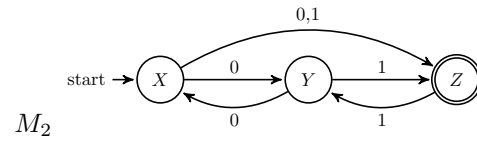
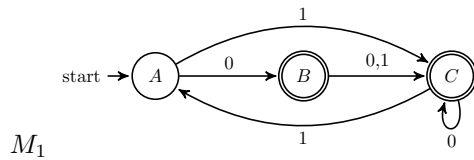


Test 3

1 Below are state diagrams for two automata, M_1 and M_2 . Let $A = L(M_1)$ and $B = L(M_2)$ be the languages recognized by these machines.



a) For each machine, circle DFA or NFA to indicate which type of machine it is.

M_1 : DFA NFA

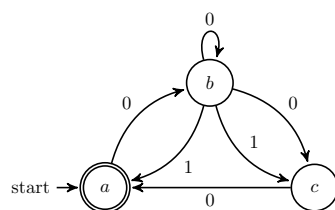
M_2 : DFA NFA

b) Complete the following table.

x	$x \in A?$ (Y or N)	$x \in B?$ (Y or N)
01		
001		
101		
0010		

c) Create a new NFA for the language $A \cup B$ using a general method that works for the union of any two languages.

2 Below is a the state diagram for an NFA N .



Draw the state diagram for a DFA that accepts the same language using an algorithm that can convert any NFA into an equivalent DFA. (You need not show unreachable states.)

3 Let L be the language that contains all and only binary strings that

- contain at least three 1s, and
- end 010.

(So $101010 \in L$ and $11010 \in L$, but $10110 \notin L$ and $10010 \notin L$.)

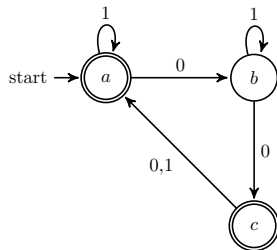
a) Design a DFA that accepts the language L . (Partial credit for an NFA.)

b) Construct a regular expression for L . (You do not have to use the algorithm that converts any DFA into a regular expression. Just come up with some correct regular expression.)

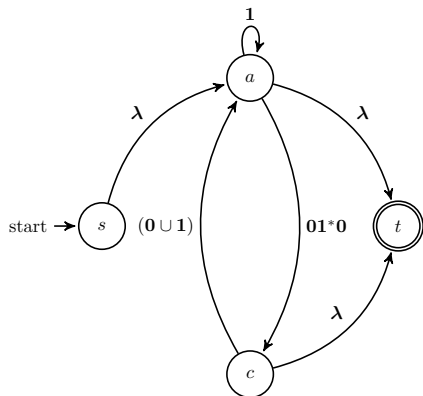
4 We learned an algorithm for converting any DFA into a regular expression. This problem asks you a few questions about that algorithm.

- a) In the first phase of the algorithm, we must modify the automaton to make sure it satisfies some properties that are necessary for the algorithm to work correctly. What are those properties?

- b) Demonstrate how to do the first phase using the DFA below.



- c) In the second phase, states are “ripped out” of a GNFA until only two states remain. Using that process, show what happens when state a is ripped out of the GNFA shown below. (You don’t have to complete the process, just do this one step.)



5 The production rules for a grammar G with start symbol S , terminals a , b , and c and non-terminals X , Y , and Z is given below:

$$\begin{array}{lcl} G : & S & \rightarrow XY \\ & X & \rightarrow a \mid YZ \\ & Y & \rightarrow b \mid ZX \\ & Z & \rightarrow c \mid XY \end{array}$$

a) Show that G generates $bacab$ (i.e., that $S \xRightarrow{*} bacab$).

b) Is G a regular grammar? Explain.

c) Is G a type 2 (context free) grammar? Explain.

6 Show that if A is a regular language, then the complement of A is also regular. (We can denote the complement of A as $\overline{A} = A^c = \{x \mid x \notin A\}$.)

_____ Space for additional work. _____